

Rotational States of Adsorbed H₂ on Fe(OH)₃ Molecule

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(Received February 21, 2020; accepted July 30, 2020; published online September 4, 2020)

Dispersion-corrected density functional (DFT-D3) and Quantum perturbation theories are used to calculate the potential energy surface (PES), rotational states and energy rotational levels of adsorbed hydrogen molecule on Fe(OH)₃. The PES is calculated as a function of vertical distance and polar and azimuthal angles. PES curves as a function of vertical distance, show that molecular hydrogen prefers to lie flat on Fe(OH)₃ with its center of mass above the Fe³⁺ site at a vertical height of 2.7 Å, and with a potential well depth of −3.062 kcal/mol. The azimuthal potential energy barrier is found to be very small and hence H₂ molecules are likely to rotate freely on Fe(OH)₃. Quantum perturbation theory calculations show that the binding energy of H₂ on Fe(OH)₃ depends on the rotational state of the molecule. Above the Fe³⁺ site of Fe(OH)₃, the para-H₂ ($J = 0$, $m = 0$) molecule has the strongest binding energy of −2.817 kcal/mol, the helicoptering ortho-H₂ molecule ($J = 1$, $m = \pm 1$) has a binding energy of −2.365 kcal/mol, and the cartwheeling ortho-H₂ molecule ($J = 1$, $m = 0$) has the weakest binding energy with a value of −1.355 kcal/mol on Fe³⁺ site.

1. Introduction

Behavior of H₂ molecules on surfaces and clusters, which acts as quantum rotors, are of fundamental importance in a wide range of processes and have attracted considerable attention, being the subject of several experimental and theoretical studies.^{1–12} According to the function for the symmetry of the total nuclear spin (I), hydrogen molecules can be classified as one of two spin isomers known as ortho (o -H₂) and para (p -H₂), i.e., o -H₂ and p -H₂, which have $I = 1$ and 0, respectively. Since protons are indistinguishable fermions, the total molecular wave function of hydrogen molecules must be anti-symmetric with respect to the permutation of the two H₂ nuclei. This antisymmetric property restricts the rotational states of ortho (o -H₂) and para (p -H₂) to odd and even rotational quantum numbers (J) respectively.^{5,13} To a good approximation, the adsorbed molecular hydrogen with $J = 0$ behaves as if it was spherically symmetric with an isotropic charge distribution, whereas the $J = 1$ molecule has an anisotropic charge distribution and can be either in the $m = 0$ or in the $m = \pm 1$ states (where m is magnetic quantum number), which are commonly referred to as “cartwheel” and “helicopter” orientations, respectively.

Systems such as H₂/LiF(001),^{6–8} H₂/MgO(001),^{9–11} H₂/NaCl(001),^{6,12} and H₂/SrTiO₃(001)¹³ have been investigated and suggested to be used as means to separate molecular hydrogen species according to their quantum rotational states through an physical adsorption–desorption process. The H₂/Fe(OH)₃ system has been of interest for some time and has come under investigation using density functional theory (DFT).⁴ According to these studies, the H₂ adsorbs over the Fe(OH)₃ cluster on top of Fe³⁺ atom at a tilt angle of 70° with respect to the cluster normal, and a potential well at 3.0 Å with a depth of around −0.4078 kcal/mol. From this, the authors⁴ conclude that the molecular hydrogen physically adsorbed on Fe(OH)₃. Although the equilibrium structure and energies are obtained for H₂/Fe(OH)₃ by using the standard DFT method, further investigations using more accurate methods, such as dispersion-corrected density functional theory (DFT-D3) method, are needed to determine adsorption properties of the H₂/Fe(OH)₃ system. Furthermore, the electric field strength and symmetry

of the Fe(OH)₃ cluster are different from other ionic surface in systems such as H₂/LiF(001),^{6–8} H₂/MgO(001),^{9–11} H₂/NaCl(001),^{6,12} and H₂/SrTiO₃(001),¹³ and hence it is expected to behave differently from these systems. Therefore, it is useful to investigate the quantum rotational states and rotational energy levels of H₂/Fe(OH)₃. To the best of our knowledge no quantum rotational states and rotational energy levels have been reported for H₂/Fe(OH)₃ using dispersion-corrected density functional theory and perturbation theory.

In this paper, we will report the results of the potential energy surface (PES) of H₂ adsorbed on the Fe(OH)₃ molecule by using DFT-D3 which includes van der Waals dispersion interactions. Furthermore, we will report results, calculated using perturbation methods, of quantum rotational states and rotational energy levels of a hydrogen molecule adsorbed on the Fe(OH)₃ cluster. This study will provide improved results and shed some light on the possibility of using Fe(OH)₃ as a means to separate hydrogen molecules via physical adsorption according to rotational states through an adsorption–desorption process.

2. Computational Details

The dispersion corrected to include van der Waals interactions were adopted through the Grimme approach, using the atom pair-wise additive scheme DFT-D3 method^{14,15} and subsequently we adopted the same approach to account for the dispersion effect in our calculations. We utilized the state of the art gradient corrected hybrid functional in Becke’s three-parameters expression with Lee, Yang, Parr exchange B3LYP,^{16–19} with the following basis sets: Hay–Wadt pseudo-potentials LANL2DZ basis sets for the (Fe) atom, as these basis sets are widely used to calculate the equilibrium geometries and spectroscopic properties of clusters, particularly in studies on clusters with heavy atoms and transition metal compounds,^{20–22} for hydrogen and oxygen atoms we utilized the cc-pVTZ basis sets (correlation-consistent polarized valence triple zeta).^{23–26} The cc-pVTZ basis sets are used successfully to simulate H₂O and H₂O₂ systems.²⁶ All DFT-D3 calculations were performed with (Gaussian09W revision D.01) software²⁷ and GaussView 5.0.9 software²⁸ to compute the optimized structures and adsorption energies for free H₂ molecule,

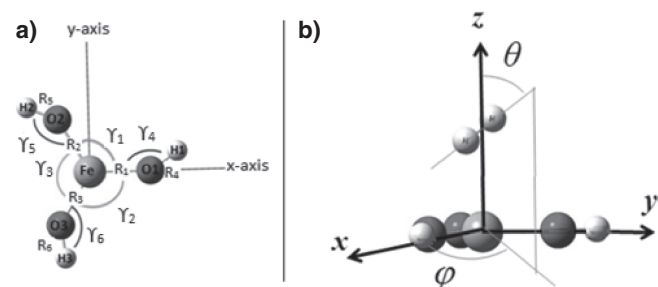


Fig. 1. (a) Top view of schematic diagram of the proposed Fe(OH)₃ structure with the bond lengths and angles: $Y_1 = 120^\circ$, $Y_2 = 120.1^\circ$, $Y_3 = 119.9^\circ$, $Y_4 = 124.6^\circ$, $Y_5 = 124.7^\circ$, $Y_6 = 124.7^\circ$, $R_1 = R_2 = R_3 = 1.797$ Å, and $R_4 = R_5 = R_6 = 0.691$ Å. (b) A view of the H₂/Fe(OH)₃ system. The origin of the coordinate system is chosen to be the position of Fe³⁺ ion in xy-plane. The z-axis is directed along the surface normal. The center of mass of H₂ is located directly above the Fe³⁺. The molecular orientation with respect to the surface normal is given by θ , and the azimuthal orientation of the H-H bond with respect to the x-axis is given by φ .

Fe(OH)₃ and H₂/Fe(OH)₃ systems. We should emphasise that the inclusion of van der Waals interactions to DFT calculations (DFT-D3) is fundamental in physical adsorption to assure correct adsorption properties. DFT-D3 has been shown to perform better with a Becke-type exchange such as B3LYP functionals.¹⁵⁾

Muhida et al.⁴⁾ using standard DFT, predicted that molecular hydrogen has a physisorption nature on Fe(OH)₃ and as a result we expect that the gas phase rotational states of the hydrogen molecule will be perturbed by the electric field provided by the ionic surface. Therefore, in this work we have applied the perturbation theory to determine the rotational states of the adsorbed hydrogen molecule on Fe(OH)₃. The molecular vibrational motion of the H₂ molecule can be neglected, as was done in the study of H₂/LiF(001),^{7,8)} H₂/NaCl(001),¹²⁾ and H₂/SrTiO₃(001)¹³⁾ systems, and hence, the adsorbed molecule is considered to be rigid (fixed bond length to $r = 0.74$ Å) and allowed only translations in the perpendicular direction along the z-axis as well as rotations around the center of mass between its two atoms. We use one Fe(OH)₃ molecule as the surface model. The atoms of the Fe(OH)₃ cluster are not allowed to move during the calculation. The coordinate system and proposed structure of Fe(OH)₃ is shown in Figs. 1(a) and 1(b). The orientation of the hydrogen molecule can be described conveniently with respect to the shown coordinate system by defining, in the standard way, a polar angle θ (tilt with respect to the surface normal) and an azimuthal angle φ (angle between the x axis and the projection of the molecular axis onto the surface model plane), while Z is the distance from the H₂ center-of mass (CM) to the Fe³⁺ ion of Fe(OH)₃. The adsorption energy or binding energy (E) of H₂ is defined here as: $E = E_T - E_s - E_{\text{gas}}$ where E_{gas} refers to the total energy of a H₂ molecule in the gas phase; E_s refers to the total energy of the Fe(OH)₃ cluster before an adsorbate of interest is introduced, and E_T refers to the total energies of both the Fe(OH)₃ cluster and the adsorbed H₂ molecule. With this definition, negative values of adsorption energy denote that the adsorbed state is more stable than the corresponding clean cluster and gas phase H₂. The calculated potential energies do not include zero point energy corrections.

Table I. Optimized geometry parameters (bonds R1 to R6 and angles Y1 to Y6) of Fe(OH)₃ surface as shown in Fig. 1. These parameters are calculated using DFT-D3/B3LYP with LANL2DZ basis sets for Fe atom and by using cc-pVTZ basis sets for hydrogen and oxygen atoms and compared with Muhida et al.⁴⁾ results.

Structure parameters	Our results using DFT-D3 and LANL2DZ/cc-pVTZ basis sets	Muhida et al. ⁴⁾
R ₁ (Å)	1.797	1.786
R ₂ (Å)	1.797	1.788
R ₃ (Å)	1.797	1.786
R ₄ (Å)	0.961	0.969
R ₅ (Å)	0.961	0.969
R ₆ (Å)	0.961	0.969
Y ₁ (deg.)	120.0	120.5
Y ₂ (deg.)	120.1	119.8
Y ₃ (deg.)	119.9	119.7
Y ₄ (deg.)	124.6	145.0
Y ₅ (deg.)	124.6	143.2
Y ₆ (deg.)	124.7	143.1

3. Results and Discussion

First, we used the DFT-D3 method to optimize the structural properties of the H₂ molecule in gas phase. The calculated bond length ($r_{\text{H-H}}$) and binding energy of isolated molecular hydrogen (E_{H_2}) are found to have values of 0.74 Å and -110.25344 kcal/mol respectively. These values compare well with the experimental values of 0.74 Å²⁹⁾ and -103.27 kcal/mol.³⁰⁾ The structure parameters and energies of Fe(OH)₃ were also optimized using DFT-D3/B3LYP/LANL2DZ(Fe)/cc-pVTZ (H,O). The calculated equilibrium structure parameters (bonds and angles) of Fe(OH)₃ are given in Table I. The calculated total binding energy of Fe(OH)₃ is found to be -220854.52 kcal/mol. Our optimized structure parameters of Fe(OH)₃ are in good agreement with published experimental and theoretical results.⁴⁾

DFT-D3 with computational level B3LYP/LANL2DZ(Fe)/cc-pVTZ(H,O) is implemented in Gaussian09W-revision-D.01 software²⁷⁾ to optimize the H₂/Fe(OH)₃ system. The initial configuration is constructed as following: the H₂ molecule is placed at a distance of about 5.5 Å above the fixed Fe(OH)₃ surface model. The center of mass of H₂ located directly above the Fe³⁺ atom. The H₂ molecule is allowed to move and rotate in the space. The atoms of the Fe(OH)₃ molecule are kept frozen and are not allowed to move during the simulation. The final configuration is shown in Fig. 1. This optimization shows that the H₂ molecule is laying flat on the Fe(OH)₃ cluster and its center of mass is on top the Fe atom at a vertical distance (Z) of 2.7 Å. This value of vertical distance is smaller than the value of 3.0 Å reported by Muhida et al.⁴⁾ Our calculated binding energy of H₂ on Fe(OH)₃ is found to be -3.062 kcal/mol. Since we are using DFT-D3, the dispersion can contribute significantly to the binding energy and because of that our calculated value of binding energy of H₂ on Fe(OH)₃ is much stronger than that reported by Muhada et al.⁴⁾ (-0.408 kcal/mol) using the standard DFT. In general, our calculated optimized structure for the H₂/Fe(OH)₃ system are in agreement with the previous results of Muhada et al.⁴⁾ The results obtained in this test allows us to pursue the next step of our investigation, namely calculation of the potential

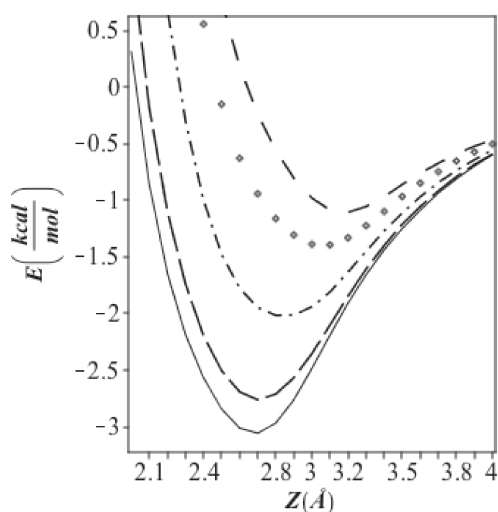


Fig. 2. Calculated potential energy curves as a function of the distance Z between the center of mass of H_2 and the Fe atom in the $Fe(OH)_3$ cluster. The space-dash, dot, dash-dot, long-dash, and line curves correspond to the following adsorption polar orientations: ($\theta = 10^\circ$), ($\theta = 30^\circ$), ($\theta = 50^\circ$), ($\theta = 70^\circ$), and ($\theta = 90^\circ$) respectively.

energy surface (PES) of H_2 molecules on the $Fe(OH)_3$ cluster as function of the height of the molecule above the surface of the cluster.

The PES as a function of (Z, θ) in the space around the optimized equilibrium position $Z = 2.7 \text{ \AA}$; is calculated. The H_2 molecule is placed on top of the fixed $Fe(OH)_3$. At each polar angle θ , the vertical position of the center of mass of the molecule is varied along z -axis directly on top of the Fe atom (the center of mass of H_2 on-top Fe site). The polar angle θ is varied from a perpendicular orientation ($\theta = 0^\circ$) to a parallel orientation ($\theta = 90^\circ$). Each time, the polar angle is increased by 10° . We consider 9 approaches for adsorption paths ($\theta = 0, 10, 20, 30, 40, 50, 60, 70, 80, 90^\circ$). The potential energy surface curves as functions of Z for $\theta = 10, 30, 50, 70, 90^\circ$ are plotted in Fig. 2. The results show that the horizontal approach leads to the lowest potential well depth. Our calculations show that the H_2 molecule has the smallest values of potential well depth when it is oriented perpendicular to the cluster (with tilt angle $\theta = 0^\circ$). Muhida et al.^{3,4} using standard DFT with-out dispersion correction, reported that for the case when $\theta = 10^\circ$, there is a potential well at $Z = 4.2 \text{ \AA}$ with a depth of around -0.1004 kcal/mol and for the case when $\theta = 70^\circ$ the potential well has a minimum (-0.408 kcal/mol) at $Z = 3.0 \text{ \AA}$. Our calculated minimum energies E and its corresponding equilibrium distances (Z) at ($\theta = 10$ and 70°) are compared to the results of Muhida et al.^{3,4} in Table II. Our calculations show that molecular hydrogen has the highest potential well depth (-1.077 kcal/mol) when it is vertical to the $Fe(OH)_3$ cluster ($\theta = 0^\circ$) at $Z = 3.19 \text{ \AA}$ and its center of mass is on top of the Fe atom. The lowest potential well depth (-3.062 kcal/mol) occurs when the hydrogen molecule is laying flat on top of $Fe(OH)_3$ cluster at $Z = 2.7 \text{ \AA}$. Therefore, the vertical position at $Z = 2.7 \text{ \AA}$ for the flat molecule orientation is considered the equilibrium configuration for the hydrogen molecule on the $Fe(OH)_3$ cluster. The position and the small depth of these potential wells are typical for physical adsorption.

In a manner similar to that used above to investigate the potential energy surface as a function of Z and θ , the effect of

Table II. The calculated energies $E_{\min}$ in (kcal/mol) of H_2 on $Fe(OH)_3$ at $Z_{\min}$ ($\text{\AA}$). $Z_{\min}$ is position of the calculated potential well measured from the center of mass of H_2 and Fe^{3+} atom. Our results is calculated by using DFT-D3/B3LYP with LANL2DZ(Fe)/cc-pVTZ(O,H) basis sets, and compared with results of Muhida et al.⁴

Polar angle in (deg.)	$E_{\min}$ (kcal/mol) This work	$Z_{\min}$ ($\text{\AA}$) This work	$E_{\min}$ (kcal/mol) Muhida et al. ⁴	$Z_{\min}$ ($\text{\AA}$) Muhida et al. ⁴
0	-1.077	3.19	—	—
10	-1.109	3.17	-0.1004	4.2
30	-1.398	3.06	—	—
50	-2.023	2.84	—	—
70	-2.761	2.71	-0.4080	3.0
90	-3.062	2.70	—	—

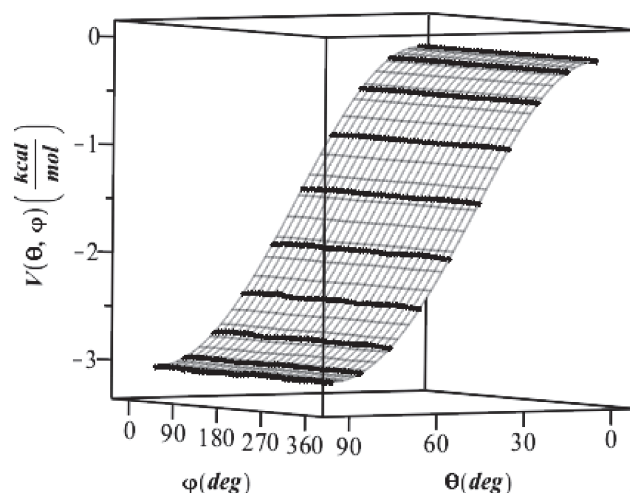


Fig. 3. The potential energy curves for H_2 adsorbed on $Fe(OH)_3$ as a function of azimuthal angle (ϕ) and polar angle (θ) at distance $2.7 \text{ \AA}$. The calculated data, are represented by dark lines. The fitting function is represented by surface-wireframe.

the azimuthal angle is investigated in Sect. 3. The center of mass of H_2 is fixed at $Z = 2.7 \text{ \AA}$ on top of the Fe atom and its polar angle is fixed at $\theta = 90^\circ$. The azimuthal angle ϕ is varied from 0 to 360° . Our results show that the influence of variation in the azimuthal angle on the potential energy surface is very small [approximately 0.6% of the adsorption energy of the H_2 on $Fe(OH)_3$] and hence there is no preferred azimuthal angle and the molecule most likely will rotate freely. In Fig. 3 we show the potential energy curve as a function of azimuthal and polar angles at $Z = 2.7 \text{ \AA}$.

To calculate the rotational states and determine which spin isomers of molecular hydrogen (*ortho* and *para*) are preferentially adsorbed over the Fe^{3+} site, rotational potential must be constructed. The potential energy surface as a function of θ and ϕ is calculated using DFT-D3 method with B3LYP/LANL2DZ(Fe)/cc-pVTZ(O,H): the molecular centre of mass is fixed at the equilibrium height of $2.7 \text{ \AA}$ directly above Fe^{3+} site, and the molecule is rotated by increments of 5° through all possible values of the polar angle θ and azimuthal angle ϕ . The calculated potential energy surface, as function of θ and ϕ , is shown in Fig. 3. In order to obtain an equation that represents the calculated rotational potential data we used the least squares fitting program provided in MAPLE V.7 [Waterloo, Maple, Maple Handbook (Maple V, release 4, 1996)]. The following function was obtained:

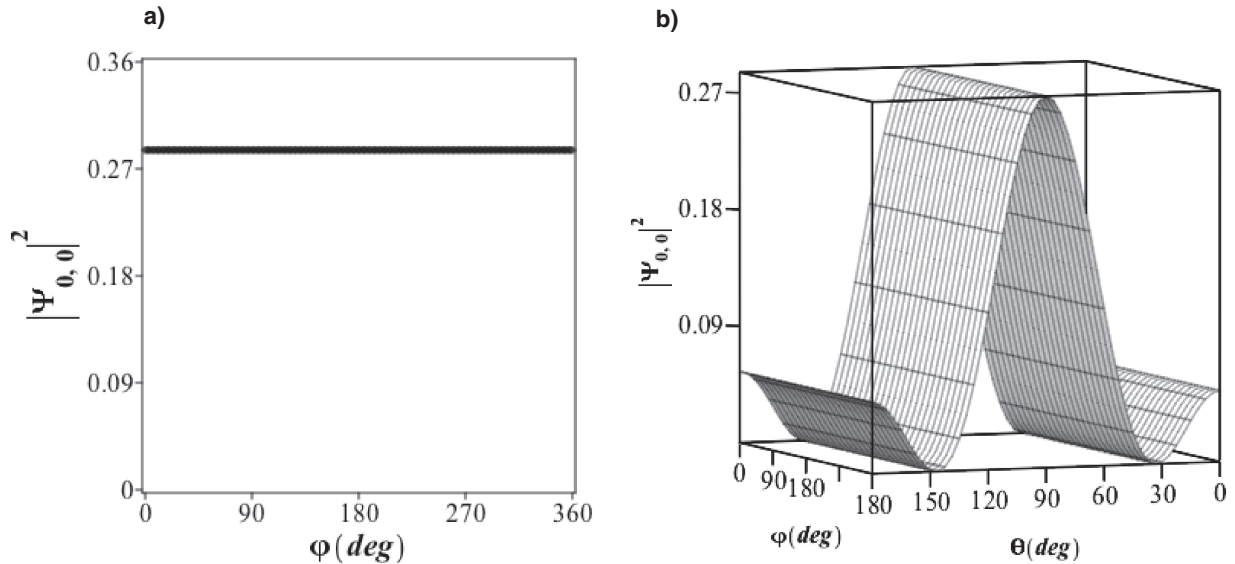


Fig. 4. (a) The angular distribution (probability density) of *para*-H₂ molecule locates over Fe³⁺ site. (b) The azimuthal distribution for $\theta = 90^\circ$ (the most probable polar orientation).

Table III. The A_n coefficients in Eq. (1), were determined using the least squares fitting method in units of kcal/mol.

A_n	A_0	A_1	A_2	A_3	A_4	A_5	A_6
Values	-3.1464	-0.0010	2.9145	0.0896	-0.0022	0.0036	0.0034

$$V(\theta, \varphi) = A_0 + A_1 \cos(6\varphi) + A_2 \cos^2(\theta) + A_3 \cos^2(2\theta) + A_4 \cos^2(\theta) \cos(6\varphi) + A_5 \cos^4(\theta) \cos(6\varphi) + A_5 \sin(2\varphi) \sin(\theta). \quad (1)$$

The A_n coefficients were determined using the least squares fitting program from Maple V. 7 and the values are listed in Table III. As done in Ref. 8, the wave functions of H₂ in different rotational state were estimated using time-independent perturbation theory to the first order correction level and may be calculated using the following formula:

$$\Psi_{Jm} \approx Y_{Jm} + \sum_{k \neq J} \frac{\langle Y_{Jm} | \mathbf{H} | Y_{km} \rangle}{E_J - E_k} Y_{km}, \quad (2)$$

where the perturbed Hamiltonian $\mathbf{H} = V(\theta, \varphi)$; J is the rotational quantum number, m is rotational magnetic quantum number, and $Y_{Jm}(\theta, \varphi)$ is the spherical harmonics functions. We used up to nine spherical harmonics ($k = 9$). From Eqs. (1) and (2) we get the following rotational wave functions for the lowest order *para* and *ortho* states.

1. For *para*-H₂ ($J = 0, m = 0$); $\Psi_{00} = 0.2821 - 0.8082Y_{20} - 0.00775Y_{40}$
2. For *ortho*-H₂ ($J = 1, m = \pm 1$); $\Psi_{11} = Y_{11} - 0.3395Y_{31} - 0.0037Y_{51}$
3. For *ortho*-H₂ ($J = 1, m = 0$); $\Psi_{10} = Y_{10} - 0.4395Y_{30} - 0.0048Y_{50}$

The probability density of *para*-H₂ in its ground state ($\Psi_{00}^* \cdot \Psi_{00} = |\Psi_{00}|^2$) is shown in Fig. 4(a) as a function of θ and φ . The probability densities distribution shows that the rotational ground state of a *para*-H₂ molecule is squashed on the surface with the highest angular probability at $\theta = 90^\circ$, i.e., the molecular axis spends most of the time parallel to the surface of the cluster. In Fig. 4(b), the azimuthal dependence of the distribution with the polar angle fixed at $\theta = 90^\circ$ is shown. The distribution is almost uniform with a very small ripple that can be neglected. In other words, the orientation of the molecular axis is largely delocalized with a very little residual preference for orientations in the $\varphi = 33, 147^\circ$ directions. Remarkably similar behavior is observed for hydrogen adsorbed on H₂/LiF(001)^{7,8} and H₂/NaCl(001)¹² systems.

The probability density of helicoptering *ortho*-H₂ ($J = 1, m = \pm 1$) is shown in Figs. 5(a) and 5(b) as a function of θ and φ . The probability distribution shows that the rotational ground state of the helicoptering *ortho*-H₂ molecule is squashed on the surface of the cluster with the highest angular probability at $\theta = 90^\circ$ and $\varphi = 0, 180, 360^\circ$. The probability distribution has lowest values at ($\varphi = 90, 270^\circ$).

The rotational state of adsorbed cartwheeling *ortho*-H₂ ($J = 1, m = 0$) suffers deformation with respect to gas phase state due the presence of the surface of the cluster. This is illustrated in Fig. 6 where the orientation distribution of the molecular axis is peaked around $\theta = 50.6$ and 129.4° . As will be discussed in this section, the cartwheeling states above the Fe³⁺ sites are energetically unfavorable.

Perturbation theory also yields the rotational energy levels of hydrogen molecule adsorbed on Fe(OH)₃, namely

$$E_{jm_j} = \frac{j(j+1)\hbar^2}{2I} + A_0 + A_2 \langle \Theta_j^{m_j} | \cos^2(\theta) | \Theta_j^{m_j} \rangle + A_3 \langle \Theta_j^{m_j} | \cos^2(2\theta) | \Theta_j^{m_j} \rangle - \frac{I[A_2 \langle \Theta_j^{m_j} | \cos^2(\theta) | \Theta_{j+2}^{m_j} \rangle + A_3 \langle \Theta_j^{m_j} | \cos^2(2\theta) | \Theta_{j+2}^{m_j} \rangle]^2}{\hbar^2(2j+3)}. \quad (3)$$

Note that:

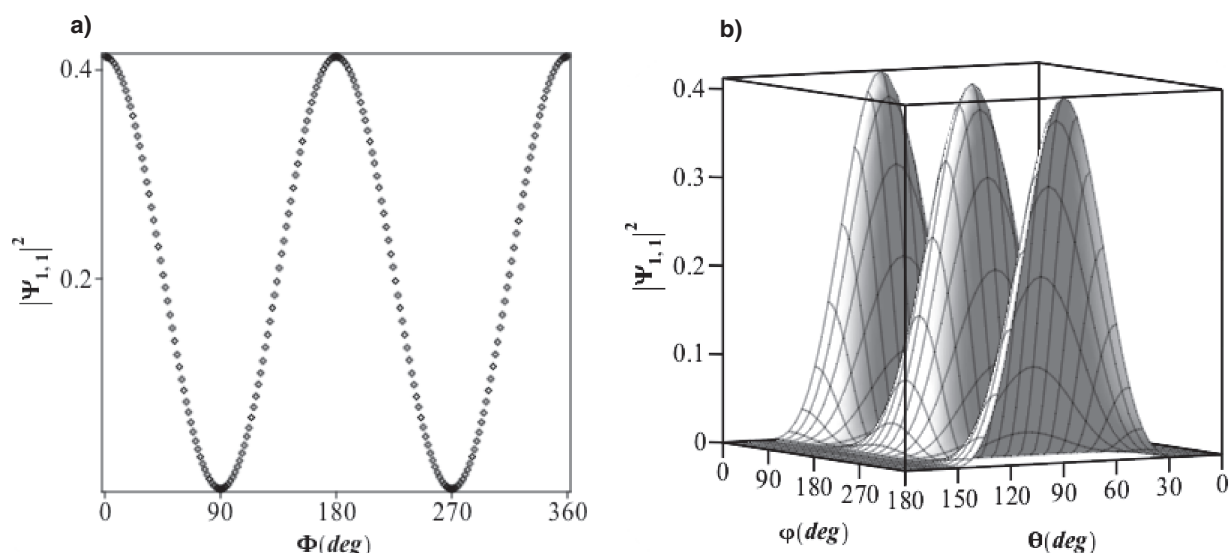


Fig. 5. (a) The angular distribution (probability density) of helicoptering *ortho*-H₂ molecule locates over Fe³⁺ site. (b) The azimuthal distribution for $\theta = 90^\circ$ (the most probable polar orientation).

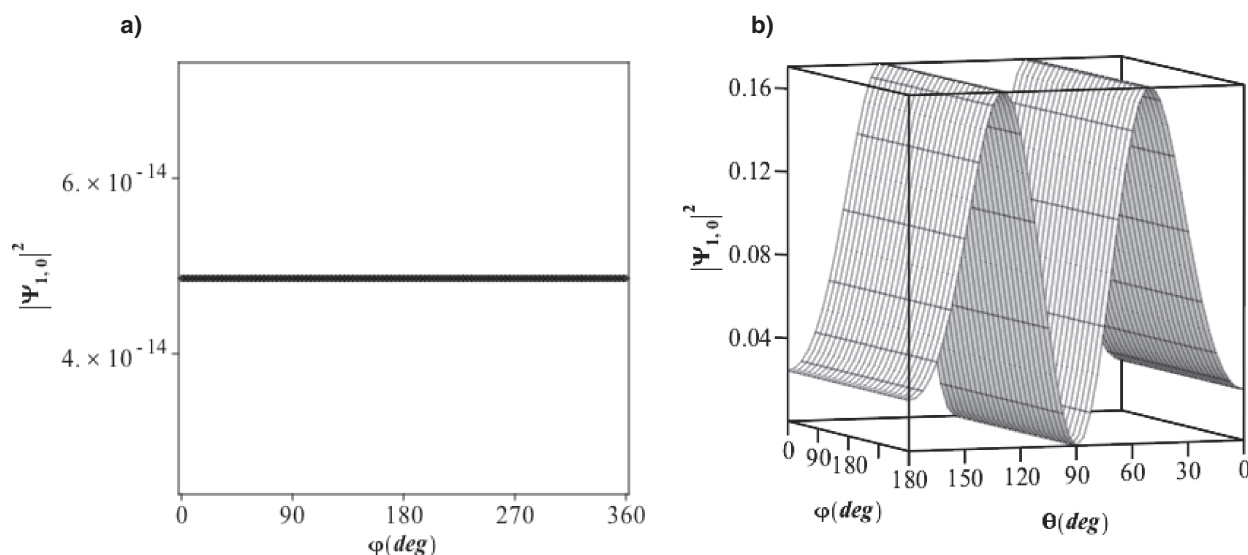


Fig. 6. The angular distribution (probability density) of cartwheeling *ortho*-H₂ molecule locates over a Fe³⁺ site.

Table IV. The calculated rotational energy levels of the of H₂ adsorbed on Fe(OH)₃.

Adsorption site	Rotational energy levels (kcal/mol)		
	para-H ₂ ($J = 0, m = 0$)	Helicoptering ortho-H ₂ ($J = 1, m = \pm 1$)	Cartwheeling ortho-H ₂ ($J = 1, m = 0$)
Fe ³⁺	-2.817	-2.365	-1.355

$$E_j^{(0)} - E_{j+2}^{(0)} = \frac{j(j+1)\hbar^2}{2I} - \frac{(j+2)(j+3)\hbar^2}{2I}$$

$$= -\frac{(2j+3)\hbar^2}{I}. \quad (4)$$

The energies of different H₂ rotational spin isomers above the Fe³⁺ site; have been calculated and are listed in Table IV. Each rotational energy level J of the hydrogen molecule adsorbed on top of the Fe³⁺ site splits into an ordered sequence of levels as shown in Fig. 7. The $|m| = J$ states

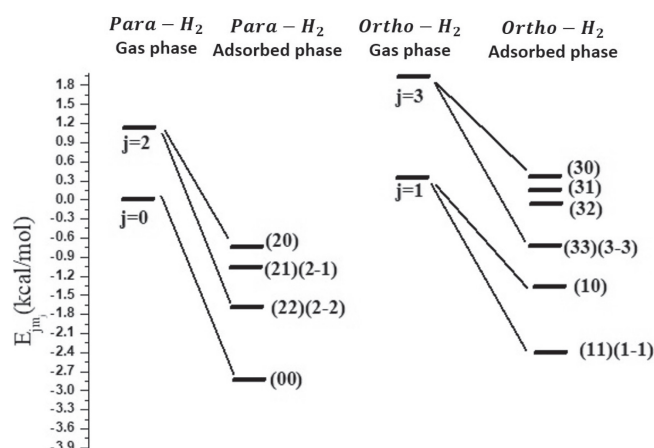


Fig. 7. The rotational energy levels of H₂ molecule in the gas phase and adsorbed phase on Fe(OH)₃ cluster above Fe³⁺ site.

have the lowest energy in the set, while when $J \neq 0$ the $m = 0$ state always has the highest energy. It is clear that in each set of states, the states ($J, m = \pm J$ including the para- H_2 with $J = 0$ state) have the lowest rotational energy and have the strongest binding energy compared to the ($J \neq 0, m = 0$) states which have the weakest binding energy. This sequence of binding energies is qualitatively similar to that reported for H_2 on LiF, MgO and NaCl surfaces.^{6–8} Above the Fe^{3+} site the para- H_2 ($J = 0, m = 0$) has the strongest binding. The cartwheeling *ortho*- H_2 ($J = 1, m = 0$) has weakest bonding energy.

4. Conclusion

Through the combination of perturbation theory calculations and Dispersion-corrected density functional theory calculations (DFT-D3/B3LYP) methods coupled with LANL2DZ basis sets for (Fe) atom and cc-pVTZ basis sets for hydrogen and oxygen atoms, the geometries, binding energies, rotational energy levels and quantum rotational states of hydrogen molecule adsorbed on the $\text{Fe}(\text{OH})_3$ have been investigated. The results show that the H_2 molecule prefers to lay parallel to the $\text{Fe}(\text{OH})_3$ cluster above the Fe^{3+} site at height of 2.7 Å. Potential barrier dependence on azimuthal orientation is very weak and the molecule is likely to rotate freely. The adsorption is predicted to be physical type (physisorption). Rotational states and energy levels calculations predict that the binding energy of the hydrogen molecule on the $\text{Fe}(\text{OH})_3$ cluster depend on the quantum rotational state of the molecule. Above the Fe^{3+} site, the para- H_2 ($J = 0, m = 0$) molecule has the strongest binding energy of -2.817 kcal/mol. The helicoptering *ortho*- H_2 molecule ($J = 1, m = \pm 1$) has a binding energy of -2.365 kcal/mol, while the Cartwheeling *ortho*- H_2 molecule ($J = 1, m = 0$) has the weakest binding energy of -1.355 kcal/mol. Our results provide for the first time detailed information about quantum rotational states and rotational energy levels of the $\text{H}_2/\text{Fe}(\text{OH})_3$ system. A molecule in the rotational state ($J = 0, m = 0$), para-hydrogen is more strongly adsorbed than the *ortho*-hydrogen. These results give insights into the possibility of using this system as a means to separate spin isomers of molecular hydrogen via physical adsorption.

Acknowledgements The authors wish to thank Misurata University (Libya) for their support.

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